A 16 GB 1024 GB/s HBM3 DRAM With Source-Synchronized Bus Design and On-Die Error Control Scheme for Enhanced RAS Features

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***Abstract*— This article proposes practical design techniques to enhance performance and reliability of 1024 GB/s high-bandwidth memory-3 (HBM3). Effective data-bus design methods are applied to transfer data from multi-bank to a data-bus with a sufficient data fetch margin. A symbol-based on-die error-correcting code (OD-ECC) to correct a 16-bit error, bounded by a sub-wordline (WL), and parallelized data-bus inversion (DBI) are implemented. Error check and scrub (ECS) mode and repair capability check (RCC) mode with an internal serial interface are designed to support system reliability, availability, and serviceability (RAS). A memory built-in self-test (MBIST) provides a unified at-speed test with programmability based on test-set units (TUs). A 16 GB HBM3 fabricated in the third generation of the 10 nm class DRAM process achieves a bandwidth up to 1024 GB/s (8 Gb/s/pin) and provides stable operation at a high temperature (e.g., 105 °C) while improving an error detection rate by 92.2%.**

***Index Terms*— 3-D-stacking memory, data-bus, DRAM, error check and scrub (ECS), high-bandwidth memory (HBM), memory built-in self-test (MBIST), on-die error-correcting code (OD-ECC), system reliability, availability, and serviceability (RAS).**

I. Introduction

HIGH-bandwidth memory (HBM) technology, which responds to higher data throughput of accelerators, such as GPUs and NPUs [1], [2], has evolved due to wide I/O interfaces and 3-D stacking techniques that are advantageous for increasing performance and capacity while maintaining a form-factor and low-power consumption [3], [4], [5], [6], [7]. As shown in Fig. 1, bandwidth and capacity have increased by 1.5–2 times every two years [4]. Due to its large capacity and a high bandwidth, HBM has been the most preferred memory technology, and it is expected that the growth of artificial intelligence (AI) and data center markets will drive continuous improvement of HBM.

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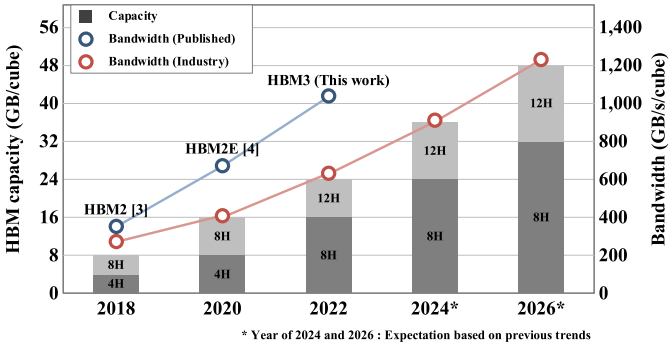


Fig. 1. HBM trend: bandwidth and capacity increase 1.5–2 times every two years.

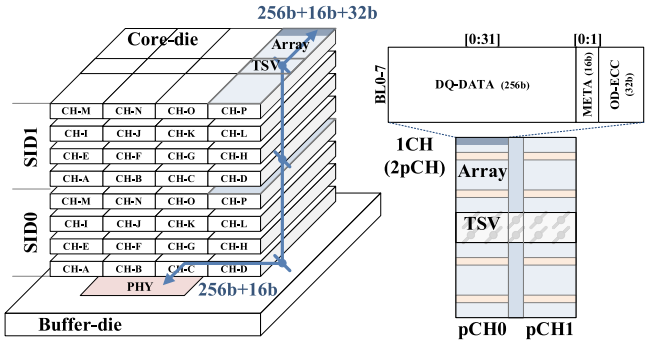


Fig. 2. 16 GB HBM3 cube with two-SID, 16-channel, and 304-bit pre-fetch data.

Fig. 2 shows a 16 GB HBM3 cube with eight core-die stacks and one buffer-die. The 16-channel is composed of a basic four-high stack, which is identified by a stack-ID (SID). Access granularity, which refers to pre-fetch size, is 304-bit including 256-bit DQ-data, 16-bit metadata, and 32-bit check-bit for on-die error-correcting code (OD-ECC), of which DQ-data and metadata are transmitted through the corresponding I/O pins. As shown in Table I, the bandwidth of HBM3 is significantly improved due to the increase in pin data rate. Therefore, this requires advanced design techniques for high-performance and stable data control.

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TABLE I
COMPARISON OF HBM2 [3], HBM2E [4], AND HBM3

Items	HBM2 [3]	HBM2E [4]	HBM3
Pin data rate	2.4 Gb/s/pin	5.0 Gb/s/pin	8.0 Gb/s/pin
Burst length	BL2 legacy mode BL4 pseudo-channel mode	BL4 pseudo-channel mode	BL8 pseudo-channel mode
The number of CH and IO	8CH, 128IO/CH 16pCH, 64IO/pCH	16pCH, 64IO/pCH	32pCH, 32IO/pCH
Access granularity (DQ only)	32B (256-bit pre-fetch)	←	←
Bandwidth	307–346 GB/s	640 GB/s	1,024 GB/s
Voltage (VDDC/VDDQ/VDDQL/VPPE)	1.2V/1.2V/−/2.5V	1.2V/1.2V/−/2.5V	1.1V/1.1V/0.4V/1.8V
Cube density	8/4/2 GB (8/4/2-high)	16/8 GB (8/4-high)	24/16 GB (12/8-high)
Core-die density	8 Gb + 1 Gb (ECC)	16 Gb + 2 Gb (ECC) + 1.5 Gb (OD-ECC)	16 Gb + 1 Gb (metadata) + 2 Gb (OD-ECC)

In addition, as its application is expanding into servers and data centers requiring server-level reliability, availability, and serviceability (RAS) standards, many challenges have been raised to HBM. To meet the server-level RAS, recent system-on-chips adopt: 1) Chipkill, which can correct any errors that occur in a single DRAM device on a DIMM [8] and 2) DDR5-Bounded Fault (DDR5-BF), which can correct multi-bit errors (MBEs) by bounding the error patterns in a DRAM and applying symbol correction code on a host [9]. Unfortunately, these methods are inefficient to be implemented in the HBM architecture. In HBM, a cache line is fetched from a single bank in a single DRAM die within the stack, so it is difficult to make a symbol bound to practical size. Therefore, this requires new approaches for the controllability of MBEs considering the HBM architecture. Moreover, in the case of DIMM, burn-in and high-speed tests are performed on memory vender sides after module mounting process. However, HBM is an environment in which additional defects can occur due to system-in-package (SiP) fabrication, even if the tests have already been performed on memory vendor sides. Therefore, the tests for HBM should be performed on host sides after assembly [4]. Furthermore, considering the thermal coupling of GPUs or ASICs during lifetime [10], a system RAS method managed after assembly is required.

This article presents a 16 GB HBM3 circuit and practical techniques considering system RAS while achieving its bandwidth up to 1024 GB/s with the third generation of the 10 nm class DRAM process: 1) we show a data-bus architecture and control scheme for achieving 8 Gb/s/pin while maintaining a form-factor and lower operating voltage; 2) present practical design methodology for OD-ECC considering the array architecture of HBM3, which is the first JEDEC HBM to apply symbol-based ECC, and error management techniques for enhancing RAS; and 3) propose an enhanced memory built-in self-test (MBIST) for an at-speed test with programmability to provide test flexibility with minimum overhead.

This article is organized as follows. Section II presents a proposed data-bus control and alignment technique to ensure a sufficient data fetch margin under reduced clock cycle time (tCK). Section III introduces configuration of the OD-ECC in HBM3 and its implementation methods considering internal data-bus inversion (DBI) computation and design-for-test (DFT) methodology. Section IV explains inter-die integration for error check and scrub (ECS) mode and repair capability check (RCC) mode designed for supporting system RAS. Section V describes a test-set unit (TU)-based programmable MBIST for an at-speed test. Section VI shows hardware measurement results of the 10 nm class 16 GB HBM3 chip, and the conclusions are drawn in Section VII.

II. HIGH-PERFORMANCE DATA-BUS DESIGN

This section presents the design of data-buses to achieve 8 Gb/s/pin in the proposed HBM3. Fig. 3 illustrates the chip and bus architecture of HBM2E [4] and HBM3. In the array region, there are DRAM cells, decoders, and I/O controllers for sensing or driving data. In the middle region, DWORD for data through-silicon-via (TSV) and AWORD for command TSV are located. To transfer data, the data-bus consists of bank group data-bus (BG-BUS), global data-bus (GBUS), and TSV.

BG-BUS, located in the array region, is a BG-BUS dedicated to each bank group. The window of BG-BUS is typically relieved by specifying tCCDL (4nCK), which means the interval between column address strobe (CAS) commands for the same bank group. The value of tCCDL is determined by considering core cycle time and OD-ECC computation time. Data of all bank groups is transmitted to TSV, at tCCDS, meaning the interval between CAS commands for any bank group.

As shown in the left side of Fig. 3, in HBM2E, GBUS, which is a GBUS to deliver data between BG-BUS and TSV in DWORD, has data transactions at tCCDS (2nCK). However, the 2nCK window of GBUS, equivalent to 1 ns at 8 Gb/s/pin, is insufficient to ensure stable bus operation and causes speed limitation. Therefore, in HBM3, alternative data line (ADL) and data-path alignment are adopted to data-bus architecture as shown in Fig. 3 (right).

A. ADL Implementation Method

The proposed HBM3 has two sets of GBUS (GBUS-1 and GBUS-2), and each GBUS is accessed alternatively by

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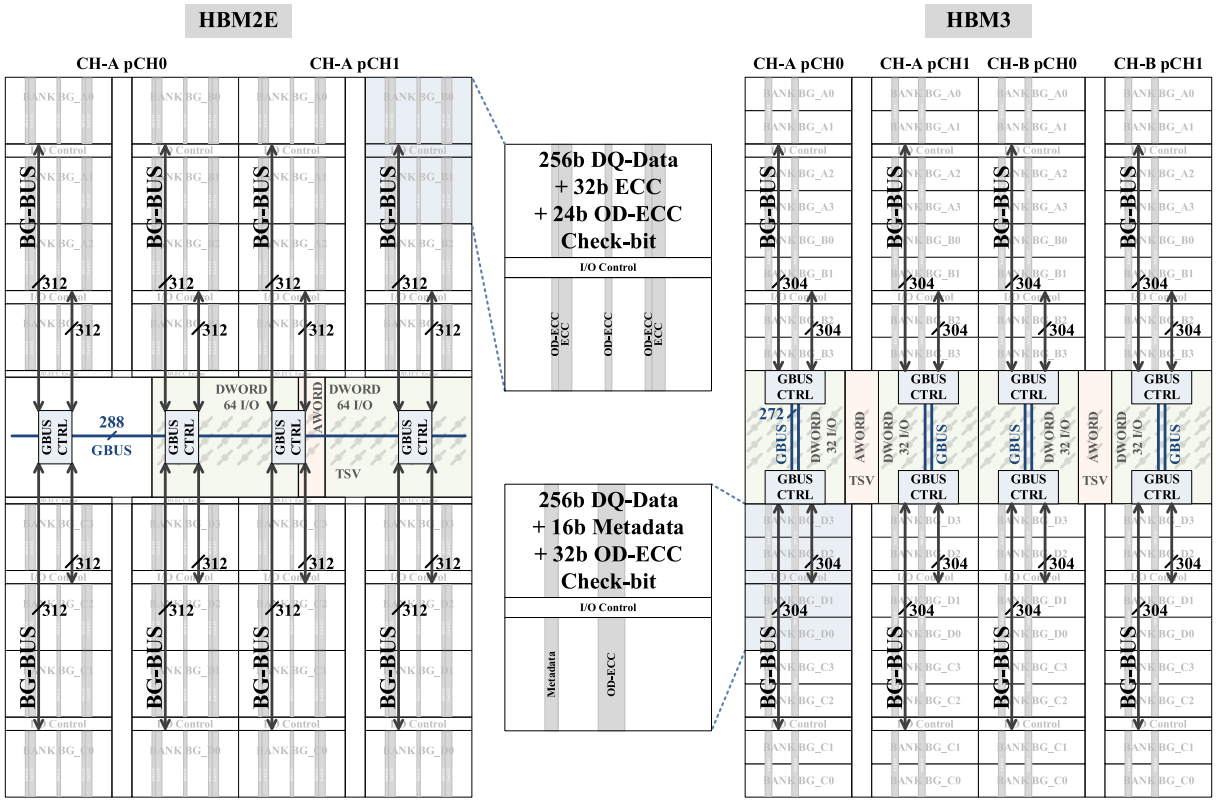


Fig. 3. Data-bus architecture of HBM2E [4] and HBM3.

every CAS command. It allows the window of each GBUS to be extended to 4nCK (2 × tCCDS), whereas that of HBM2E is 2nCK (tCCDS). Since GBUS-SEL (GBUS-SEL-1 and GBUS-SEL-2), which is a control signal that drives data to the corresponding GBUS, is activated in turn according to the CAS command that comes every tCCDS, the order of the GBUS is determined alternately. Therefore, the window of each GBUS is extended to 4nCK (2 × tCCDS). The ADL timing diagram is shown in Fig. 4(a).

In particular, during read operation, since the data, transmitted to GBUS, come from all BG-BUS of a pseudo channel, it is inevitably vulnerable to skew. Therefore, to fetch the skewed data with a stable margin, GBUS-SEL should be delivered in a path that can replica the local variation of the read data-path, composed of a read command (RD-CMD) and BG-BUS. In the proposed method, GBUS-SEL, which is initially issued along with RD-CMD to track the read data-path of each top and bottom array region, respectively, determines the order of GBUS, whereas in the conventional method, bus ordering logic determines the order by referring to RD-CMD of all bank groups. Thus, the fetch margin variation is mitigated by source synchronizing and tracking method. Moreover, the area and latency overhead caused by referencing RD-CMD of all bank groups [6] is reduced. The conventional and proposed methods for determining the order of GBUS are shown in Fig. 4(b).

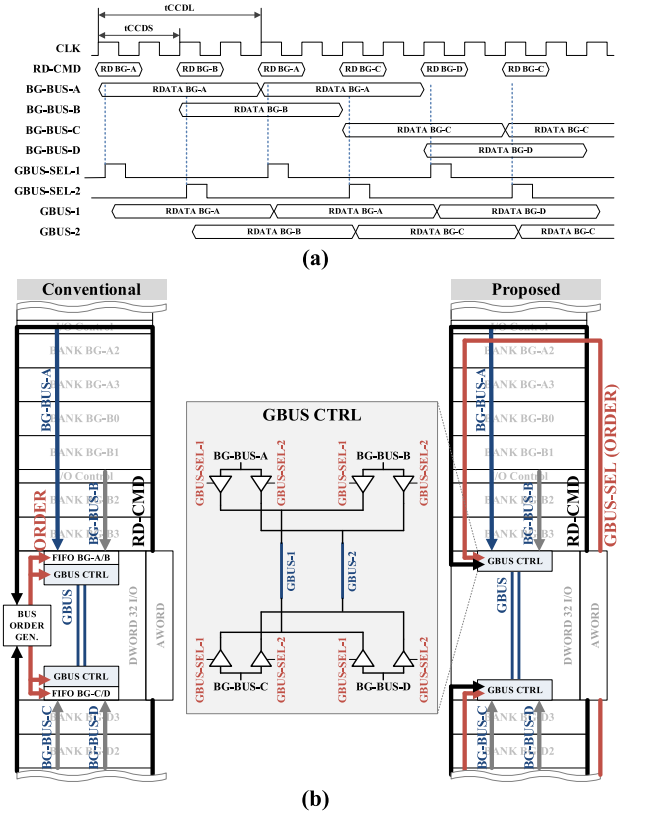


Fig. 4. (a) ADL timing diagram. (b) Implementation method of nonsynchronized GBUS control and proposed source-synchronized ADL.

B. Data-Path Alignment Layout

As shown in Fig. 3, the proposed HBM3 adopts vertical alignment of data-buses on the core-die with TSV-PHY horizontal expansion alignment on the buffer-die, in contrast to that of HBM2E [4]. BG-BUS and GBUS of the core-die

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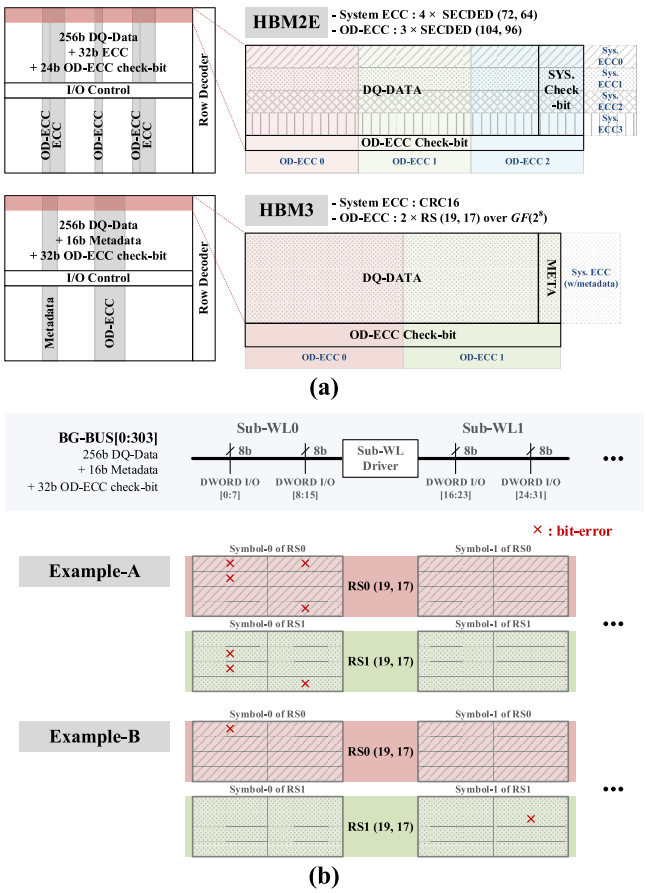


Fig. 5. (a) OD-ECC configuration of HBM2E [4] and HBM3. (b) Examples of correctable error.

is connected in alignment by matching DWORD to the corresponding array region of each pseudo-channel. Consequently, all pseudo-channels have the same bus architecture, enabling a modular design for each and optimizing the data window of GBUS to achieve high-speed data transactions. The proposed layout strategy is favorable for core-die, but it requires more resources for buffer-die: bus connection window extension using a double number of buses and associated logic is adopted to allow the signals of all channels to be transmitted between TSV and PHY at high frequency (HF). This is cost-effective considering the HBM structure combined with one buffer-die and eight core-dies.

III. OD-ECC Configuration and Implementation

This section introduces an efficient ECC configuration considering the HBM architecture and its practical implementation. HBM2E uses 32-bit for system ECC and 24-bit for OD-ECC check-bit, but only single error correction double error detection (SECDED) is applied to each ECC as shown in Fig. 5(a) (top). Although the combination of SECDEDs can improve the correctability of single-bit errors (SBEs) [4], the SECDEDs have a limitation of not being able to cope with MBEs.

However, in HBM3, the ECC configuration is optimized by considering system RAS, while minimizing check-bit

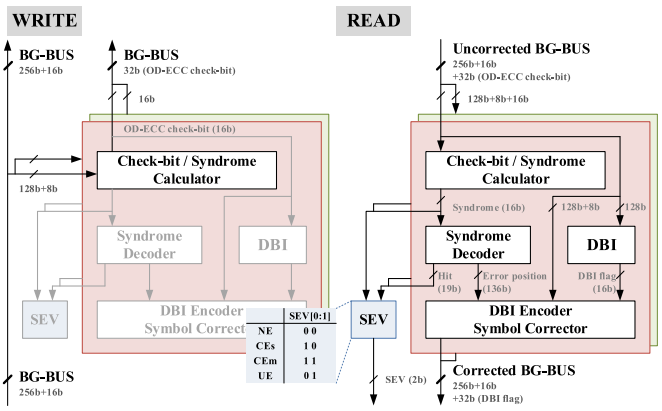


Fig. 6. Block diagram of OD-ECC during write and read operation.

resources. Since the boundaries of errors due to DRAM faults are highly dependent on DRAM architecture, error is corrected on a DRAM side using 32-bit for OD-ECC check-bit. On the other hand, robust error detection is performed on a host side using 16-bit metadata to observe errors and take action to prevent system failures considering an overall environment. Furthermore, the transparency provided by OD-ECC also contributes to improving an error detection rate.

A. Symbol-Based OD-ECC

The data of BG-BUS, consisting of 256-bit DQ-data, 16-bit metadata, and 32-bit for OD-ECC check-bit, is divided into two ECC units, and each can correct one symbol (8-bit) by applying (19, 17) Reed–Solomon (RS) code over GF(2^8), respectively, as shown in Fig. 5(a) (bottom).

Since a wordline (WL) is physically divided into sub-WLs in HBM3, as illustrated in Fig. 5(b) (top), if a sub-WL or its component has a fault, the resulting error is confined to a bounded 16-bit. To correct a bounded 16-bit error, each ECC symbol is formed to correspond to each error. Therefore, symbol-based OD-ECC can correct a bounded 16-bit error caused by a sub-WL fault as shown in the example-A of Fig. 5(b). In addition, random SBEs: one SBE per ECC, respectively, are also correctable as shown in the example-B of Fig. 5(b).

Fig. 6 shows OD-ECC operation. During write operation, 32-bit for OD-ECC check-bit is computed and stored in DRAM cells along with write data. Contrarily, during read operation, 32-bit syndrome is computed and decoded to 272-bit error position, which corrects 256-bit DQ-data and 16-bit metadata.

B. Improvement of Error Detection Rate

In HBM3, to achieve a high error detection rate, 16-bit metadata can be used as error detection code, such as cyclic redundancy check (CRC), on a host side. Additionally, an error detection rate is further improved by referring to severity levels, which are real-time transparency provided by OD-ECC. Severity levels, transferred by the SEV pins, contain the following information: 1) no error (NE); 2) corrected SBE

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TABLE II
ERROR DETECTION RATE DEPENDING ON CRC AND SEV

System ECC	w/o CRC	w/ CRC-8	w/ CRC-16
w/o real-time SEV	–	99.6094%	99.9985%
w/ real-time SEV	92.1875%	99.9695%	99.9999%

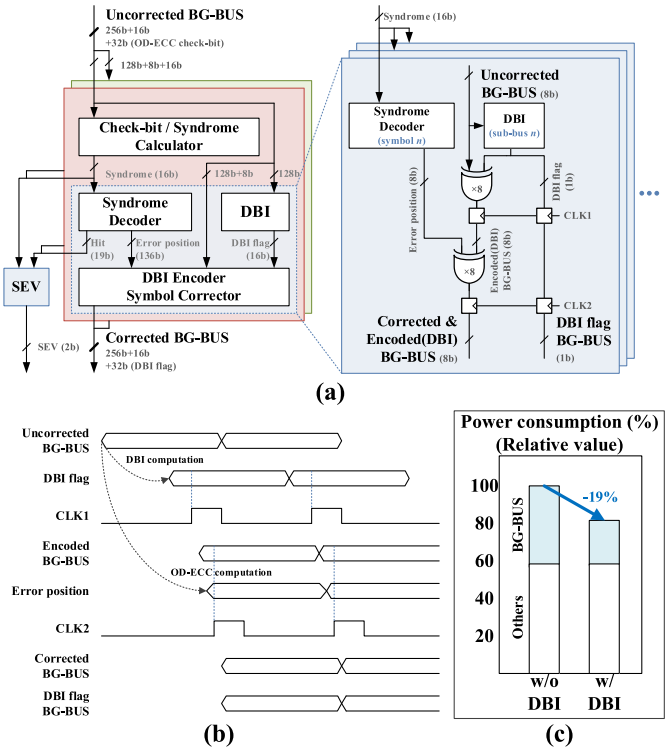


Fig. 7. (a) Block diagram. (b) Timing diagram of DBI computation parallelized with OD-ECC. (c) Power reduction of BG-BUS with DBI.

(CEs); 3) corrected MBE (CEm); or 4) uncorrectable error (UE) as shown in the table of Fig. 6. Thus, the OD-ECC decoder computes a severity level as well as error position during read operation. Using both CRC and severity levels improves an error detection rate by 92.2% compared to using only CRC, as shown in Table II.

C. Parallelized DBI

DBI, a commonly used approach to reduce the power of data-buses, is adopted to BG-BUS, which switching consumes 22%–42% of the power of the core-die. A DBI flag is added to each 8-bit sub-bus unit to indicate whether the sub-bus should be inverted to minimize switching activity. As shown in Fig. 7(a), by matching the sub-bus (8-bit) unit to the one symbol (8-bit) unit of OD-ECC, hardware logic is shared and DBI and OD-ECC is computed in a parallel manner. Hence, the proposed sharing scheme reduces the area overhead as well as the computation time during read operation. The timing diagram is shown in Fig. 7(b). As a result, by encoding BG-BUS with the DBI flag, the power of the core-die is reduced by 19% in the case of the maximum switching activity, as shown in Fig. 7(c).

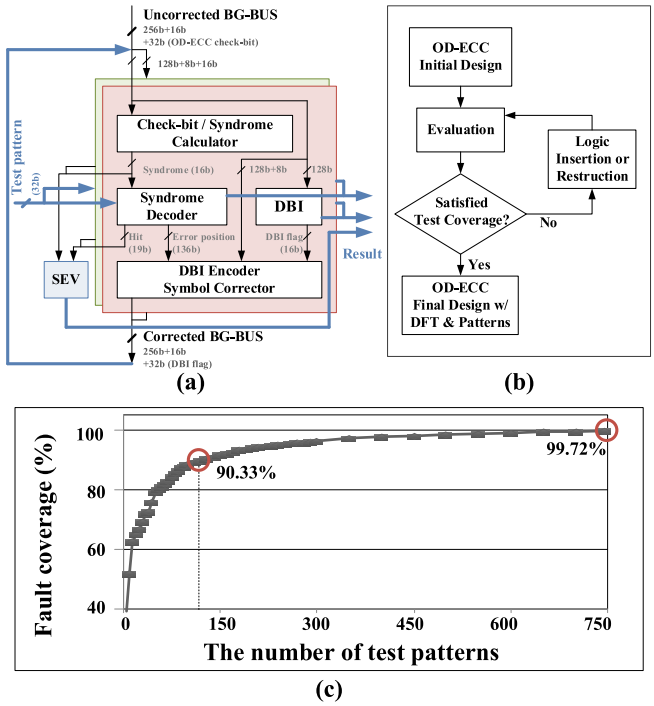


Fig. 8. (a) Block diagram. (b) Operation sequence of DFT for OD-ECC. (c) Fault coverage of OD-ECC logic depending on the number of test patterns.

D. Pattern-Aware DFT

As OD-ECC plays a significant role in system RAS, it needs to be designed to be testable with controllability and observability, which are essential elements of DFT [11]. However, due to its pattern dependency for computing check-bit (32-bit) and syndrome (32-bit), OD-ECC logic should be tested with sufficient coverage, that requires more test pins and longer test time, representing test cost.

The proposed OD-ECC DFT logic is implemented simply considering the limited number of pins in an initial test before assembly. As shown in Fig. 8(a), a 32-bit test pattern is input of the syndrome decoder. The 32-bit initial pattern is converted to 304-bit BG-BUS and entered into the syndrome calculator. By comparing the input pattern with the computed syndrome, the results of the fault detection test are obtained.

In addition, the pattern-aware DFT logic optimization method is applied, as shown in Fig. 8(b), to find the optimal set of test patterns with enough coverage within a short test time. The OD-ECC logic is efficiently isolated from other data transaction logic using fault analysis of automatic test pattern generation (ATPG). A given OD-ECC logic is analyzed to compute the test coverage and updated through logic insertion or restructure. After several iterations, an optimal DFT design and a set of patterns, considering controllability and observability, are obtained. Fig. 8(c) shows the fault coverage depending on the number of test patterns. Note that 90.33% and 99.72% of the fault coverage is achieved within 130 and 750 patterns, respectively.

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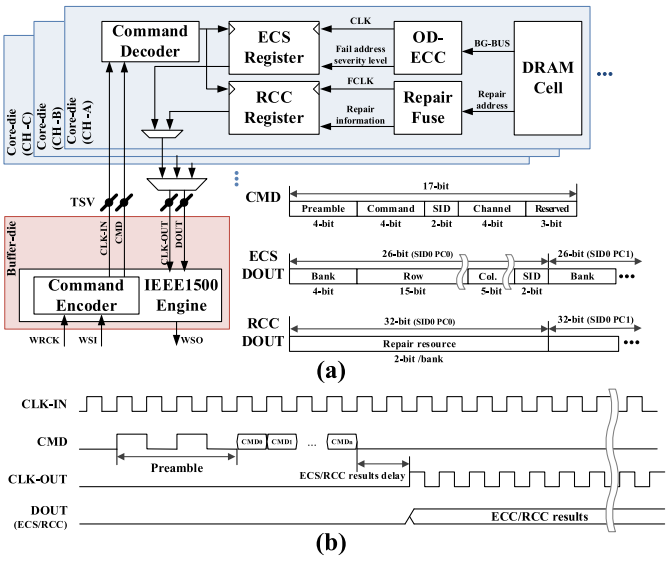


Fig. 9. (a) Block diagram and protocol of ECS and RCC. (b) Proposed internal serial interface for die-to-die communication.

IV. Error Management Techniques

To support system RAS techniques, such as page retirement [12] or post package repair (PPR) [13], HBM3 provides ECS and RCC enabling error management during operation.

A. Error Check and Scrub

ECS mode, which provides the transparency as described in the JEDEC standard [13], uses OD-ECC and operates in the background during refresh periods. It allows the DRAM to internally read, detect errors, correct errors, and write back corrected data on cells. It is performed periodically to scrub all cells of an HBM.

ECS, in HBM3, is integrated as follows: 1) if an error is detected, the error log, such as an address and severity level of the error, is stored in the ECS register in the core-die; 2) a host requests access to the log using wrapper serial input (WSI) which is IEEE1500 input; and 3) this request is transferred to the core-dies TSV, and the error log is delivered to the buffer-die to be transferred to wrapper serial output (WSO) which is IEEE1500 output. In order to transmit a request and data between the core-die and buffer-die with a limited number of TSVs, an internal serial interface is proposed, as shown in Fig. 9(a). TSVs for CMD and DOUT are assigned for the interaction between the IEEE1500 engine in the buffer-die and the ECS registers in the core-dies. Thus, the request is transferred with encoded 4-bit command and 4-bit preamble through CMD TSV, then the serialized error information is delivered through DOUT TSV accordingly. Moreover, TSVs for CLK-IN and CLK-OUT are assigned to synchronize the data between dies. The timing diagram is depicted in Fig. 9(b).

B. Repair Capability Check

In HBM3, if an error is detected during operation, a host can decide whether to repair the erroneous row or process page retirement. The proposed HBM3 supports RCC mode

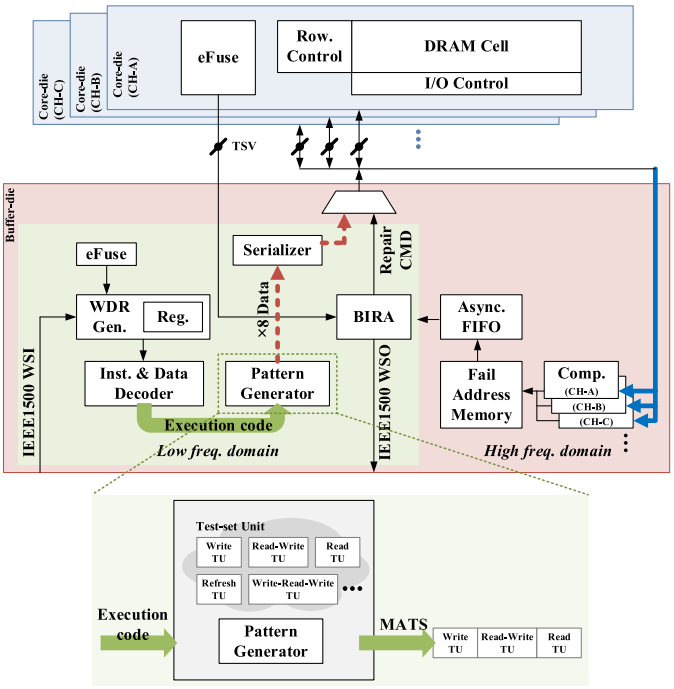


Fig. 10. Block diagram of proposed programmable MBIST.

to check the number of repair resources remaining after being used in initial evaluations. A host can use not only the resource specified in the JEDEC standard but also the resource reported as remaining by the RCC, but only the former can be used in HBM2E. This allows a host to efficiently manage recovery resources and minimize the progress of page retirement that leads to reduced memory capacity. RCC internal serial interface is similar to ECS: if a host requests access to the RCC register to check repair capability, the stored value representing the number of available resource is delivered through DOUT TSV.

V. Memory Built-In Self-Test

HBM3 also has an MBIST function, using IEEE1500 protocol, to screen manufacturing defects that occurs after SiP fabrication.

A. Unified At-Speed Test

HBM3 provides a unified at-speed test engine considering efficient power and area, while HBM2E separates low-frequency (LF) engine for cell-test and HF engine for peripheral-test. Thus, a globally asynchronous locally synchronous (GALS) approach [14] is adopted to efficiently convert between the LF test engine, which generates test patterns, and the HF transactions, which test DRAM cells and peripheral logic as in actual operating frequency. During write operation (highlighted as the red dashed line in Fig. 10), patterns for test are generated in the LF domain, and then converted to the HF domain using the serializer. In addition, data are expanded with duplication and modification functions defined by user-defined options and then delivered to the core-die. Conversely, during read operation (highlighted as the blue bold line in

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TABLE III
KEY FEATURE SUMMARY

Items	HBM2E [4]	HBM3
Process	2nd generation of 10 nm class DRAM	3rd generation of 10 nm class DRAM
Cube density	16/8 GB (8/4-high)	24/16 GB (12/8-high)
Pin data rate	5.0 Gb/s/pin	8.0 Gb/s/pin
Bandwidth	640 GB/s	1,024 GB/s
Core-die density	16 Gb + 2 Gb (ECC) + 1.5 Gb (OD-ECC)	16 Gb + 1 Gb (metadata) + 2 Gb (OD-ECC)
Correction capability	1-bit error	16-bit error (bounded by a sub-WL)
MBIST cell and core test	LF-test w/ fixed patterns	United HF-test w/ TU-based flexible patterns
MBIST peripheral test	HF-test w/ fixed patterns	

Fig. 10), data corresponding to the test result from the core-die is compared to the expected data in the HF domain, then the fail addresses are stored in the asynchronous FIFO to change frequency domain. Thus, the result is converted to the LF domain for built-in repair analysis (BIRA), which provides a method to repair the fail addresses on a host side. Therefore, the pattern generator is operated in the LF domain for minimizing power and area overhead, whereas DRAM cells and peripheral logic is tested in the HF domain for reflecting real data transactions.

B. Programmable Test

High programmability requires overhead in terms of logic complexity and setup cost. The proposed MBIST in HBM3 improves programmability while minimizing overhead by using a TU-based pattern generator operated in the LF domain and eFuse for one-time programmability. Commands for wrapper data register (WDR) are generated with IEEE1500 WSI and eFuse, and translated into execution code for pattern generator as shown in Fig. 10. Several TU elements, including a sequence of instructions, are assembled by pattern generator. For example, the modified algorithmic test sequence (MATS) [15] pattern of MBIST is generated by combining the write TU, read-write TU, and read TU as shown in Fig. 10 (bottom). Moreover, the cost is further reduced using eFuse, which simplifies setup, such as ac parameters and retention time, allowing the HBM to be tested in an actual operating environment. Therefore, the proposed MBIST not only efficiently implements a flexible sequence of the instructions using TUs, but also minimizes unnecessary setup using eFuse.

VI. HARDWARE IMPLEMENTATION AND MEASUREMENT RESULTS

A 16 GB HBM3 is implemented in the third generation of the 10-nm class DRAM process and 3-D chip stacking technology. Fig. 11 shows the chip microphotographs of the fabricated 16 Gb DRAM core-die [as shown in Fig. 11(a)] and buffer-die [as shown in Fig. 11(b)]. Cell-less high-performance transistors are used for the buffer-die. The fabricated 16 GB HBM3 consists of one buffer-die and eight-high core-die stack, and is designed to be expandable up to 12-high stack. Table III summarizes key features, including comparison with the previous HBM2E [4]. The micro pillar grid array (MPGA)

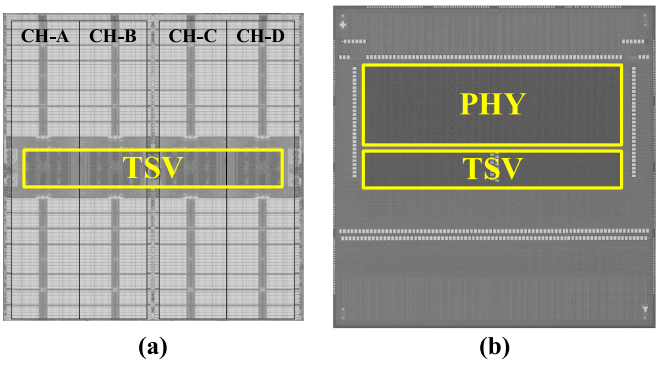


Fig. 11. Chip microphotographs of (a) core-die and (b) buffer-die.

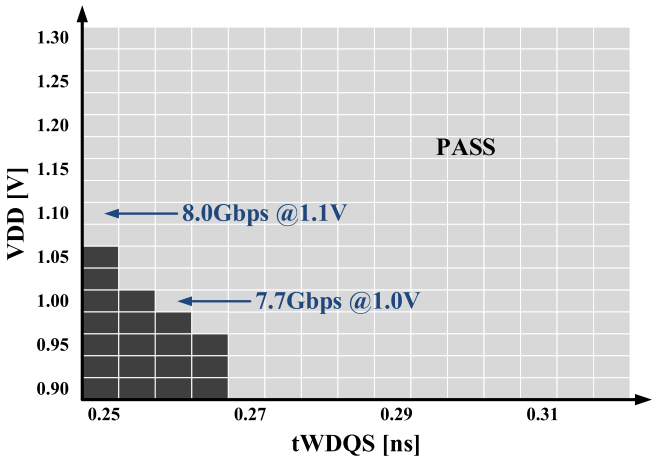


Fig. 12. Measured tWDQS shmoo.

cube size of HBM3 is 11 × 11 mm and the cube density is 16 or 24 GB, depending on the number of core-die stack. On the core-die, 16 Gb DQ-data, 1 Gb metadata, and 2 Gb OD-ECC check-bit DRAM cells are implemented. Fig. 12 shows that the wide operating VDD ranges from 0.9 to 1.3 V at 105 °C and the measured tWDQS at 1.1 V is 0.25 ns which corresponds to 8 Gb/s/pin.

VII. CONCLUSION

Several circuits and design techniques of the 16 GB HBM3 have been presented for enhancing bandwidth and supporting RAS. The proposed methods for improving data-bus transaction speed, including ADL and data-bus alignment, achieve the tWDQS of 0.25 ns corresponding to 8 Gb/s/pin and 1024 GB/s

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per cube. The OD-ECC, in HBM3, corrects bounded 16-bit errors to overcome the limitation of the conventional HBM, which has insufficient correctability due to the cache line structure. In addition, it improves an error detection rate by 92.2% by providing real-time transparency. The parallelized DBI scheme reduces the power of the core-die by 19% in the worst switching activity case, and the pattern-aware DFT achieve 99.7% fault coverage of OD-ECC logic. The ECS and RCC, designed to support system RAS, efficiently report DRAM cell scrubbing log and repair resource capability using the proposed internal serial interface. The MBIST TUs generate flexible test patterns to improve test coverage after SiP assembly while supporting a unified at-speed test.

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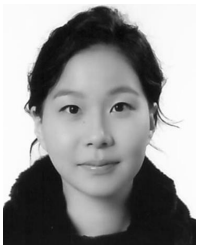
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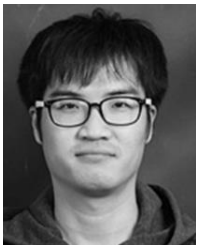
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